Chapter 4. Some Applications of Fourier Series

Isoperimetry, equidistribution, nowhere differentiable functions, heat

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4.1 The isoperimetric inequality

Motivation. Among closed curves of given length the circle encloses the largest area. Writing the curve's coordinates as Fourier series turns length and area into sums of squares of coefficients, and Parseval's identity reduces the inequality to $|n| \le n^2$.

Curves. A **parametrized curve** is a map $\gamma : [a,b] \to \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$; it is **simple** if injective on $[a,b)$, **closed** if $\gamma(a) = \gamma(b)$, and of class C^1 (with closed curves $(b-a)$-periodic). The **curve** Γ is the image. Its **length** is $\ell = \int_a^b |\gamma'(s)|\, ds$, independent of C^1 reparametrizations. Every curve with $\gamma' \neq 0$ can be parametrized by **arc-length**, $|\gamma'(s)| = 1$, and then $\ell = b - a$. For a simple closed C^1 curve enclosing the region Ω, Green's theorem gives the **area**

$$\mathcal{A} = \frac{1}{2}\Big|\int_\Gamma (x\,dy - y\,dx)\Big| = \frac{1}{2}\Big|\int_a^b (x(s)y'(s) - y(s)x'(s))ds\Big|,$$

which we take as the definition of the enclosed area.

Theorem 1.1 (Isoperimetric inequality)

If Γ is a simple closed C^1 curve in $\mathbb{R}^2$ of length ℓ enclosing area $\mathcal{A}$, then $\mathcal{A} \le \frac{\ell^2}{4\pi}$, with equality iff Γ is a circle.

Proof.

Step 1 (normalization). Rescaling $(x,y) \mapsto \delta(x,y)$ multiplies ℓ by δ and $\mathcal{A}$ by δ^2, so we may take $\ell = 2\pi$ and show $\mathcal{A} \le \pi$. Parametrize by arc-length on $[0, 2\pi]$, so $x'(s)^2 + y'(s)^2 = 1$ and

$$\frac{1}{2\pi}\int_0^{2\pi} (|x'(s)|^2 + |y'(s)|^2)ds = 1.$$

Step 2 (Fourier coefficients). Let $x(s) \sim \sum a_n e^{ins}$, $y(s) \sim \sum b_n e^{ins}$. Then $x'(s) \sim \sum a_n in e^{ins}$, $y' \sim \sum b_n in e^{ins}$ (integration by parts, periodic C^1). Parseval gives

$$\sum_n |n|^2(|a_n|^2 + |b_n|^2) = 1.$$

Since x, y are real, $\overline{a_n} = a_{-n}$, $\overline{b_n} = b_{-n}$. By Lemma 1.5 of Chapter 3,

$$\mathcal{A} = \frac{1}{2}\Big|\int_0^{2\pi} (xy' - yx')\, ds\Big| = \pi\Big|\sum_n n(a_n\overline{b_n} - b_n\overline{a_n})\Big|.$$

Step 3 (the inequality). $|a_n\overline{b_n} - b_n\overline{a_n}| \le 2|a_n||b_n| \le |a_n|^2 + |b_n|^2$ and $|n| \le |n|^2$, so

$$\mathcal{A} \le \pi\sum_n |n|(|a_n|^2 + |b_n|^2) \le \pi\sum_n |n|^2(|a_n|^2 + |b_n|^2) = \pi.$$

Step 4 (equality). If $\mathcal{A} = \pi$, then $|n| = |n|^2$ wherever $|a_n|^2 + |b_n|^2 \neq 0$, so $a_n = b_n = 0$ for $|n| \ge 2$, and $x(s) = a_{-1}e^{-is} + a_0 + a_1 e^{is}$, $y(s) = b_{-1}e^{-is} + b_0 + b_1 e^{is}$. With $\overline{a_1} = a_{-1}$, $\overline{b_1} = b_{-1}$, the identity $2(|a_1|^2 + |b_1|^2) = 1$ gives $|a_1|^2 + |b_1|^2 = \frac{1}{2}$, and equality in $2|a_1\overline{b_1} - b_1\overline{a_1}| \le \dots$ requires $2|a_1||b_1| = |a_1|^2 + |b_1|^2$, i.e. $|a_1| = |b_1| = \frac{1}{2}$, and $|a_1\overline{b_1} - \overline{a_1}b_1| = 2|a_1||b_1|$. Write $a_1 = \frac{1}{2}e^{i\alpha}$, $b_1 = \frac{1}{2}e^{i\beta}$; then $|e^{i(\alpha-\beta)} - e^{-i(\alpha-\beta)}| = 2$, so $|\sin(\alpha-\beta)| = 1$, $\alpha - \beta = k\pi/2$ with k odd. Hence $x(s) = a_0 + \cos(\alpha + s)$, $y(s) = b_0 \pm \sin(\alpha + s)$: a circle of radius 1. Conversely the circle gives equality. ∎

4.2 Weyl's equidistribution theorem

Motivation. For irrational γ the fractional parts $\langle n\gamma\rangle$ fill $[0,1)$ uniformly. Testing against exponentials, the average of $e^{2\pi ikn\gamma}$ over $n \le N$ is a geometric sum that tends to 0; trigonometric polynomials approximate continuous and then Riemann integrable functions.

For $x \in \mathbb{R}$, $[x]$ is the integer part and $\langle x\rangle = x - [x] \in [0,1)$ the **fractional part**; $x = y \bmod \mathbb{Z}$ if $x - y \in \mathbb{Z}$. For γ rational, $\langle n\gamma\rangle$ takes finitely many values; for γ irrational the $\langle n\gamma\rangle$ are distinct (if $\langle n_1\gamma\rangle = \langle n_2\gamma\rangle$, then $(n_1 - n_2)\gamma \in \mathbb{Z}$).

Definition (Equidistribution)

A sequence $\xi_1, \xi_2, \dots$ in $[0,1)$ is **equidistributed** if for every interval $(a,b) \subseteq [0,1)$,

$$\lim_{N\to\infty} \frac{\#\{1 \le n \le N : \xi_n \in (a,b)\}}{N} = b - a.$$

Theorem 2.1 (Weyl)

If γ is irrational, then $\langle\gamma\rangle, \langle 2\gamma\rangle, \langle 3\gamma\rangle, \dots$ is equidistributed in $[0,1)$. In particular $\{\langle n\gamma\rangle\}$ is dense in $[0,1)$ (**Kronecker's theorem**).

Let $\chi_{(a,b)}$ be the 1-periodic extension of the characteristic function of $(a,b) \subseteq [0,1)$. Then $\#\{n \le N : \langle n\gamma\rangle \in (a,b)\} = \sum_{n=1}^{N} \chi_{(a,b)}(n\gamma)$, and the theorem says

$$\frac{1}{N}\sum_{n=1}^{N} \chi_{(a,b)}(n\gamma) \to \int_0^1 \chi_{(a,b)}(x)\,dx. \tag{4.1}$$

Lemma 2.2

If f is continuous, 1-periodic, and γ is irrational, then $\frac{1}{N}\sum_{n=1}^{N} f(n\gamma) \to \int_0^1 f(x)\,dx$.

Proof.

Step 1 (exponentials). For $f = 1$ both sides are 1. For $f(x) = e^{2\pi ikx}$, $k \neq 0$: $\int_0^1 f = 0$, and with $e^{2\pi ik\gamma} \neq 1$ (γ irrational),

$$\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma} = \frac{e^{2\pi ik\gamma}}{N}\frac{1 - e^{2\pi ikN\gamma}}{1 - e^{2\pi ik\gamma}}, \qquad \left|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma}\right| \le \frac{2}{N|1 - e^{2\pi ik\gamma}|} \to 0.$$

By linearity the lemma holds for trigonometric polynomials.

Step 2 (approximation). Given $\epsilon > 0$, Corollary 5.4 of Chapter 2 gives a trigonometric polynomial P with $|f - P| < \epsilon/3$. Then $\left|\frac{1}{N}\sum f(n\gamma) - \frac{1}{N}\sum P(n\gamma)\right| < \epsilon/3$, $\left|\int f - \int P\right| < \epsilon/3$, and $\left|\frac{1}{N}\sum P(n\gamma) - \int P\right| < \epsilon/3$ for N large. ∎

Corollary 2.3

The conclusion of Lemma 2.2 holds for every 1-periodic f that is Riemann integrable on $[0,1]$.

Proof. Let f be real. Given $\epsilon > 0$, choose a partition with upper and lower step functions $f_U \ge f \ge f_L$ and $\int(f_U - f_L) < \epsilon$. Modifying near the partition points (as in Lemma 3.2 of Chapter 2, keeping $f_\pm$ above/below), we get continuous 1-periodic $f_+ \ge f \ge f_-$ with $\int(f_+ - f_-) < 2\epsilon$. By Lemma 2.2,

$$\limsup \frac{1}{N}\sum f(n\gamma) \le \lim \frac{1}{N}\sum f_+(n\gamma) = \int f_+ \le \int f + 2\epsilon,$$

and similarly $\liminf \geq \int f - 2\epsilon$. ∎

Proof of Theorem 2.1. Apply Corollary 2.3 to $f = \chi_{(a,b)}$, which gives (4.1). ∎

Weyl's criterion (one direction proved above, the other an exercise in the book): $\{\xi_n\} \subseteq [0,1)$ is equidistributed iff $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$. For example $\langle n^2\gamma \rangle$ is equidistributed for irrational γ. *Geometric interpretation:* a light ray reflected inside a square mirror box traces a closed periodic path if the slope is rational and a dense (indeed equidistributed) path if it is irrational.

4.3 A continuous but nowhere differentiable function

Motivation. Lacunary Fourier series $\sum 2^{-n\alpha} e^{i2^n x}$ converge uniformly but oscillate too wildly to be differentiable anywhere. Delayed Cesàro means isolate a single lacunary term, whose derivative is huge, whereas differentiability would force all such derivatives to be $O(\log N)$.

Theorem 3.1 (Weierstrass)

For $0 < \alpha < 1$, $f_\alpha(x) = f(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i2^n x}$ is continuous but nowhere differentiable.

Continuity is clear: the series converges absolutely and uniformly since $\sum 2^{-n\alpha} < \infty$. For nowhere differentiability we use the **delayed means**

$$\triangle_N(f) = 2\sigma_{2N}(f) - \sigma_N(f).$$

Facts. (a) Since $\sigma_M(f) = \sum_{|n|<M} (1 - \frac{|n|}{M}) a_n e^{inx}$, the coefficient of $a_n e^{inx}$ in $\triangle_N(f)$ is $2(1 - \frac{|n|}{2N}) - (1 - \frac{|n|}{N}) = 1$ for $|n| \leq N$ and $2 - \frac{|n|}{N}$ for $N < |n| < 2N$:

$$\triangle_N(f) = S_N(f) + \sum_{N<|n|<2N} \Big(2 - \frac{|n|}{N}\Big) a_n e^{inx}.$$

(b) For lacunary f (frequencies 2^k only) and $N = 2^{n-1}$, no frequency lies strictly between N and $2N$, so $\triangle_N(f) = S_N(f)$.

Lemma 3.2

If g is continuous and differentiable at x_0, then $\sigma_N(g)'(x_0) = O(\log N)$, hence $\triangle_N(g)'(x_0) = O(\log N)$.

Proof.

Step 1. $\sigma_N(g)'(x_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N'(t) g(x_0 - t)\, dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F_N'(t)(g(x_0 - t) - g(x_0))dt$, since $\int F_N' = 0$ (F_N periodic). Differentiability at x_0 and boundedness of g give $|g(x_0 - t) - g(x_0)| \leq C|t|$.

Step 2 (bounds for F_N'). $F_N = \sum_{|n|<N}(1 - \frac{|n|}{N}) e^{int}$ gives $|F_N'(t)| \leq \sum_{|n|<N} |n| \leq N^2$. And from $F_N(t) = \frac{\sin^2(Nt/2)}{N \sin^2(t/2)}$, [Supplement] differentiating,

$$|F_N'(t)| \leq \frac{|\sin(Nt/2)\cos(Nt/2)|}{\sin^2(t/2)} + \frac{\sin^2(Nt/2)|\cos(t/2)|}{N|\sin(t/2)|^3} \leq \frac{A}{t^2} + \frac{A}{N|t|^3} \leq \frac{2A}{t^2} \quad (|t| \geq 1/N).$$

Step 3. Hence $|\sigma_N(g)'(x_0)| \leq C\int_{|t|\leq 1/N} N^2|t|\, dt + C\int_{1/N\leq|t|\leq\pi} \frac{2A}{|t|}\, dt = O(1) + O(\log N)$. Finally $\triangle_N(g)' = 2\sigma_{2N}(g)' - \sigma_N(g)'$. ∎

Lemma 3.3

If $2N = 2^n$, then $\triangle_{2N}(f) - \triangle_N(f) = 2^{-n\alpha} e^{i2^n x}$.

Proof. By Fact (b), $\triangle_{2N}(f) = S_{2N}(f)$ and $\triangle_N(f) = S_N(f)$, and the only frequency in $(N, 2N]$ is $2N = 2^n$. ∎

Proof of Theorem 3.1. Suppose f is differentiable at x_0. By Lemma 3.3, $\triangle_{2N}(f)'(x_0) - \triangle_N(f)'(x_0) = i2^{n(1-\alpha)} e^{i2^n x_0}$, of modulus $2^{n(1-\alpha)} = (2N)^{1-\alpha}$. By Lemma 3.2 it is $O(\log N)$. But $N^{1-\alpha}/\log N \to \infty$ since $\alpha < 1$: a contradiction. ∎

4.4 The heat equation on the circle

Motivation. On a ring the temperature evolves by $\partial_t u = \partial_x^2 u$. Separating variables gives $u = \sum a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = f * H_t$: high frequencies are damped extremely fast, and the heat kernel is a good kernel.

Model the ring by the circle with angle $\theta = 2\pi x$, $x \in [0, 1)$. After rescaling time the temperature satisfies $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with $u(x, 0) = f(x)$. Separating $u = A(x)B(t)$ gives $\frac{B'}{B} = \frac{A''}{A} = \lambda$, and periodicity forces $\lambda = -4\pi^2 n^2$, $A = e^{\pm 2\pi i n x}$, $B = e^{-4\pi^2 n^2 t}$. Superposing,

$$u(x, t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}, \qquad a_n = \hat{f}(n).$$

For integrable f the a_n are bounded, the series converges with all derivatives for $t > 0$, and u solves the equation. Moreover $u(\cdot, t) \to f$ in mean square as $t \to 0$ (Parseval). In convolution form $u(x, t) = (f * H_t)(x)$ with the **heat kernel** of the circle

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x}.$$

$\{H_t\}_{t>0}$ is a family of good kernels as $t \to 0$ (Chapter 5, Corollary 3.4), so $u(x, t) \to f(x)$ at every point of continuity, uniformly if f is continuous. Unlike the Poisson kernel, H_t is not given by an elementary closed formula; its positivity and concentration follow from Poisson summation (Chapter 5).

4.5 Selected exercises

Motivation. Four exercises from the book: the converse half of Weyl's criterion and a uniform version of it, a natural sequence that fails to be equidistributed, and the mean-square boundary behaviour of the heat equation stated above.

Exercise 4.5 (Powers of the golden ratio)

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\gamma_n = \langle \varphi^n \rangle$. Prove that $\{\gamma_n\}$ is not equidistributed in $[0, 1)$.

Solution.

Step 1. Let $\psi = \frac{1-\sqrt{5}}{2}$. Both φ and ψ are roots of $x^2 = x + 1$, so $U_n = \varphi^n + \psi^n$ satisfies $U_{n+1} = U_n + U_{n-1}$, with $U_0 = 2$, $U_1 = 1$. By induction every U_n is an integer.

Step 2. $\psi \approx -0.618$, so $0 < |\psi| < 1$ and $\varphi^n = U_n - \psi^n$. If n is odd, $-\psi^n = |\psi|^n \in (0, 1)$, so $\gamma_n = |\psi|^n$. If n is even, $-\psi^n \in (-1, 0)$, so $\gamma_n = 1 - |\psi|^n$.

Step 3. Choose n_0 with $|\psi|^{n_0} < \frac{1}{4}$. For $n \geq n_0$, $\gamma_n \in [0, \frac{1}{4}) \cup (\frac{3}{4}, 1)$. Hence

$$\frac{\#\{1 \leq n \leq N : \gamma_n \in (\frac{1}{4}, \frac{3}{4})\}}{N} \leq \frac{n_0}{N} \to 0 \neq \frac{1}{2},$$

so the sequence is not equidistributed. ∎

Exercise 4.7 (Weyl's criterion, converse direction)

Let $\{\xi_n\} \subseteq [0, 1)$ be equidistributed. Prove that $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$.

Solution. Write $\#_N(E) = \#\{1 \leq n \leq N : \xi_n \in E\}$.

Step 1 (single points are negligible). Let $c \in [0, 1)$. If $c = 0$: $\#_N(\{0\}) \leq N - \#_N((0, 1))$, and $\#_N((0, 1))/N \to 1$. If $c > 0$: for small $\delta > 0$, $\#_N(\{c\}) \leq \#_N((c - \delta, c + \delta))$, whose ratio to N tends to 2δ. In both cases $\#_N(\{c\})/N \to 0$.

Step 2 (continuous f). Let f be real and continuous on $[0, 1]$, $|f| \leq B$, and $\epsilon > 0$. By uniform continuity choose $0 = x_0 < x_1 < \cdots < x_M = 1$ with $|f(x) - f(x_j)| < \epsilon$ on $[x_{j-1}, x_j]$. Every ξ_n lies in some (x_{j-1}, x_j) or equals some x_j with $j < M$, so

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) - \sum_{j=1}^{M} f(x_j)\frac{\#_N((x_{j-1}, x_j))}{N}\Big| \leq \epsilon + B\sum_{j=0}^{M-1}\frac{\#_N(\{x_j\})}{N}.$$

Step 3. As $N \to \infty$ the second sum tends to $\sum_j f(x_j)(x_j - x_{j-1})$, a Riemann sum within ϵ of $\int_0^1 f$, and the error term tends to ϵ by Step 1. So $\limsup_N \big|\frac{1}{N}\sum f(\xi_n) - \int_0^1 f\big| \leq 2\epsilon$ for every ϵ, i.e. $\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f$.

Step 4. Apply Step 3 to $\cos(2\pi k x)$ and $\sin(2\pi k x)$, whose integrals over $[0, 1]$ vanish for $k \neq 0$. ∎

Exercise 4.10 (Uniform averages along an equidistributed sequence)

Let $\{\xi_n\}$ be equidistributed in $[0, 1)$ and f continuous and 1-periodic with $\int_0^1 f = 0$. Prove that $\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n) \to 0$ uniformly in x.

Solution.

Step 1 (exponentials). For $k \neq 0$,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k(x+\xi_n)}\Big| = \Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n}\Big| \to 0$$

by Exercise 4.7, and the left side does not depend on x. By linearity the claim holds uniformly in x for every trigonometric polynomial Q with zero constant term.

Step 2 (approximation). Let $\epsilon > 0$. By Corollary 5.4 (rescaled to period 1) there is a trigonometric polynomial P with $\sup|f - P| < \epsilon$. Its constant term $c_0 = \int_0^1 P$ satisfies $|c_0| = |\int_0^1 (P - f)| < \epsilon$, so $Q = P - c_0$ has zero constant term and $\sup|f - Q| < 2\epsilon$.

Step 3. For all x,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n)\Big| \leq 2\epsilon + \Big|\frac{1}{N}\sum_{n=1}^{N} Q(x + \xi_n)\Big| < 3\epsilon$$

once $N \geq N_0$, with N_0 independent of x by Step 1. ∎

Exercise 4.11 (Mean-square initial values for the heat equation)

Let f be integrable on $[0,1]$ (1-periodic) and $u(x,t) = (f * H_t)(x)$. Prove that $\int_0^1 |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0^+$.

Solution.

Step 1. For $t > 0$, $u(\cdot,t) = \sum_n a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}$ converges uniformly ($|a_n| \le \sup|f|$), so its Fourier coefficients are $a_n e^{-4\pi^2 n^2 t}$ and those of $u(\cdot,t) - f$ are $a_n(e^{-4\pi^2 n^2 t} - 1)$.

Step 2. Parseval (Theorem 1.3, rescaled to period 1) for the integrable function $u(\cdot,t) - f$ gives

$$\int_0^1 |u(x,t) - f(x)|^2 dx = \sum_n |a_n|^2 (1 - e^{-4\pi^2 n^2 t})^2.$$

Step 3. Each factor $(1 - e^{-4\pi^2 n^2 t})^2$ lies in $[0,1]$ and tends to 0 as $t \to 0$. Given $\epsilon > 0$, choose M with $\sum_{|n|>M} |a_n|^2 < \epsilon$ (possible since $\sum |a_n|^2 = \int_0^1 |f|^2 < \infty$). Then the sum is at most $\sum_{|n|\le M} |a_n|^2 (1 - e^{-4\pi^2 M^2 t})^2 + \epsilon$, which is $< 2\epsilon$ for t small. ∎